

Qualitative data

Example 1: Consider the data about Sex of 10 students

Sex	M	F	M	M	F	M	F	M	M	M
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- Make a frequency distribution, relative frequency and % frequency of the above data and interpret your results? Make an appropriate graph?

Example 2: Suppose we have also collected data of Sections of these 10 students as

Sex	M	F	M	M	F	M	F	M	M	M
Section	A	A	A	B	B	B	A	B	A	B

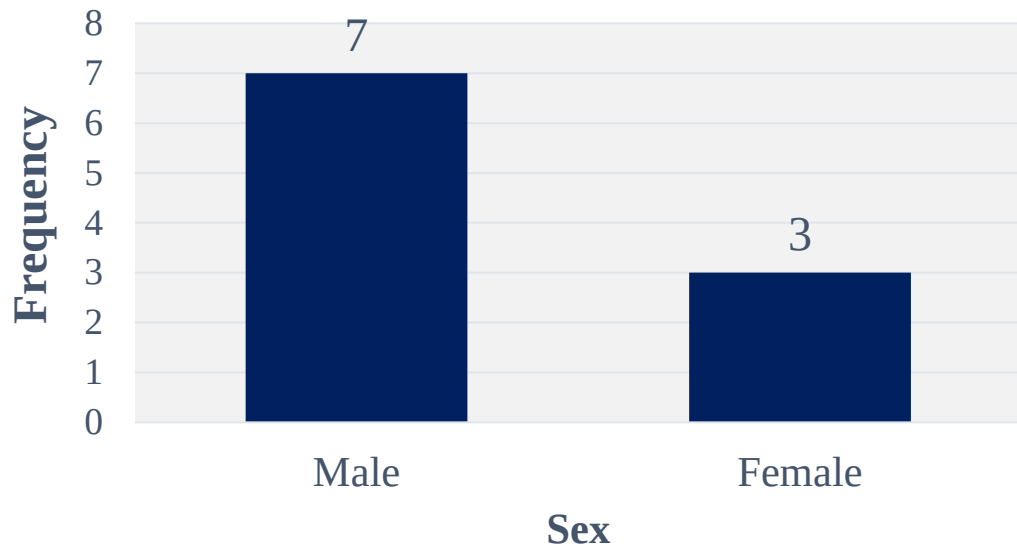
- Construct the Cross tabulation of the above data and interpret your results? Also make an appropriate graph?

Solution

Example 1

Sex	f	Relative freq	% freq
Male	7	0.7	70
Female	3	0.3	30
Total	10	1.0	100

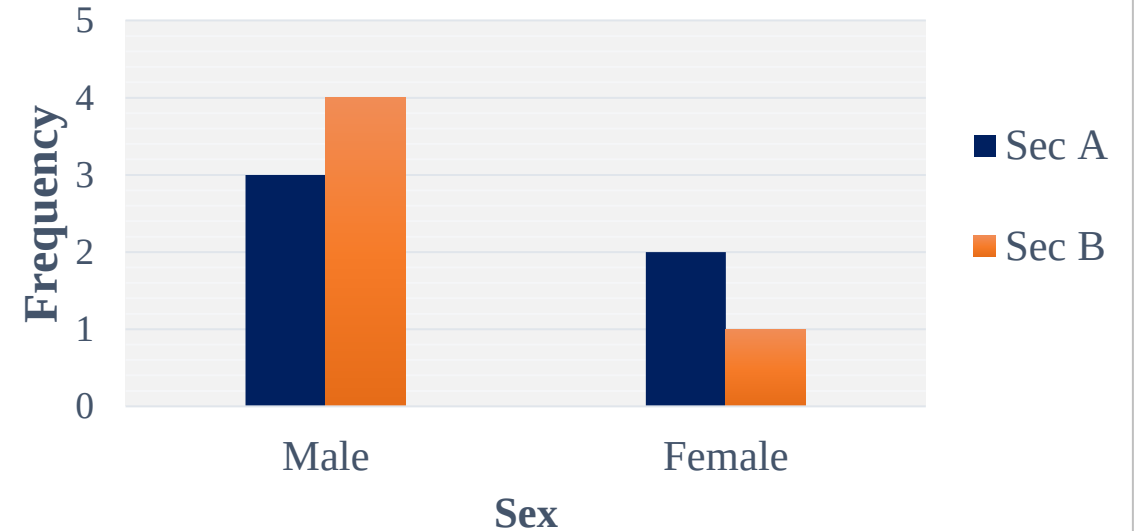
Bar Chart



Example 2

Sex	Sec A	Sec B	Total
Male	3	4	7
Female	2	1	3
Total	5	5	10

Multiple Bar chart

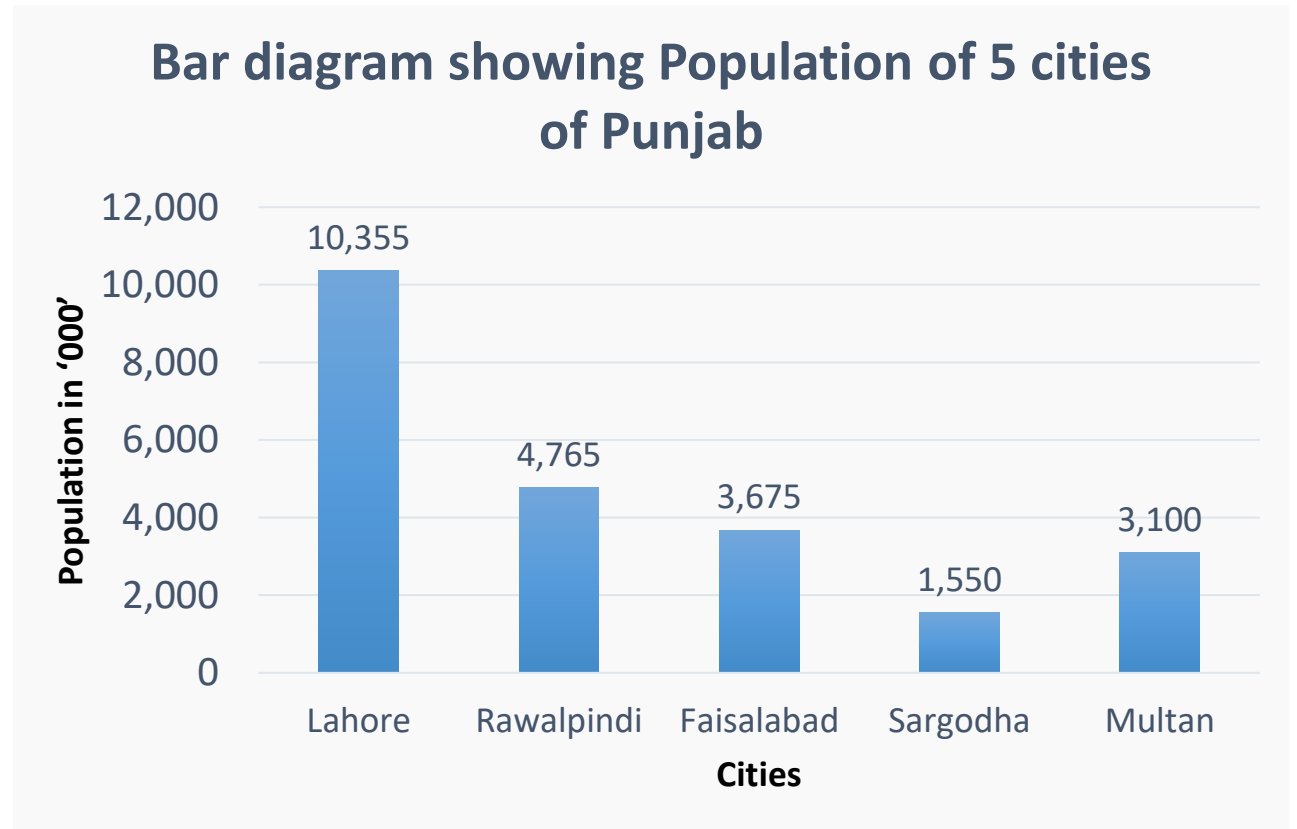


Simple Bar Chart

- A **bar chart** is a type of **chart** which shows the values of different categories of data as rectangular **bars** with different lengths.

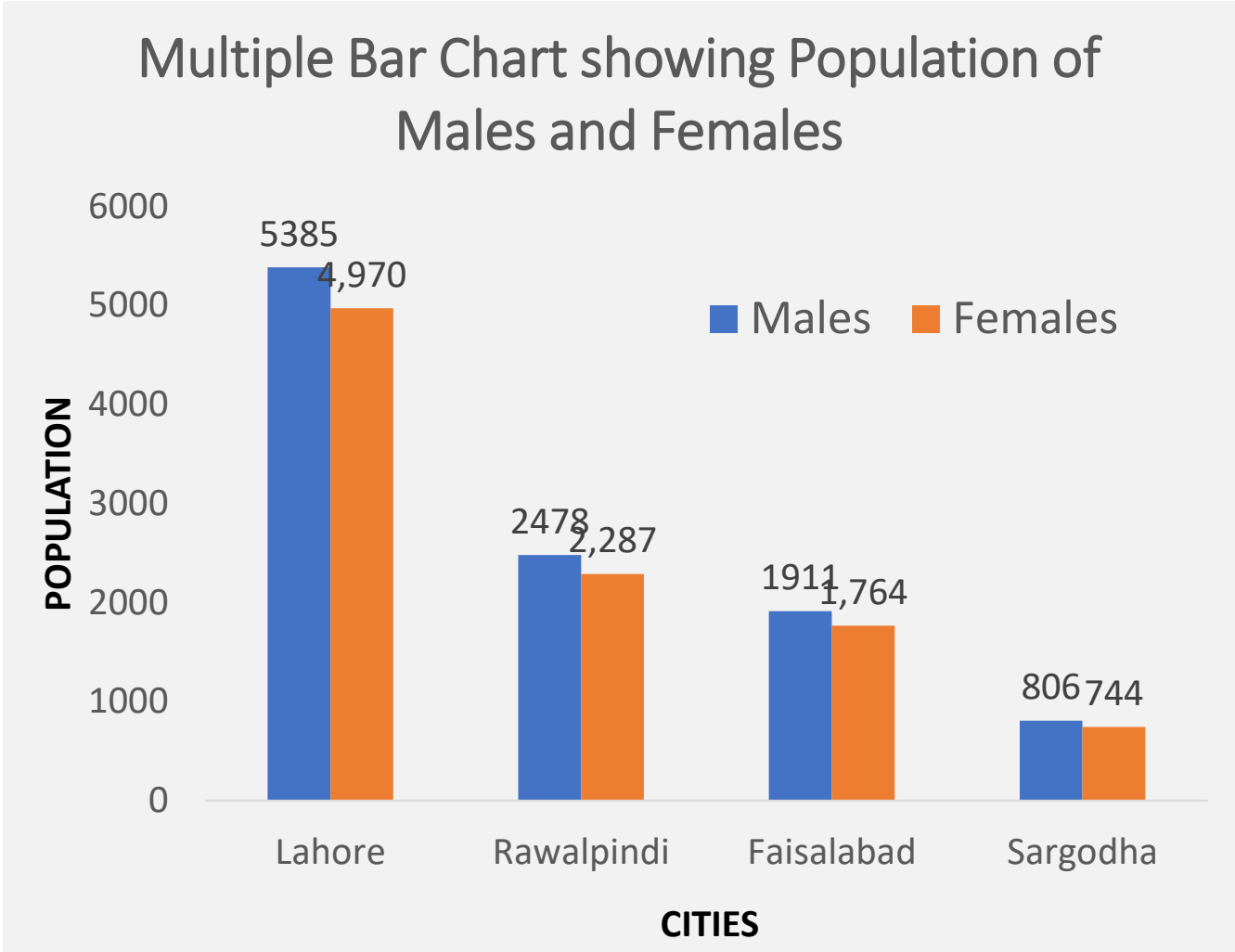
Example: Draw a Simple Bar Chart to represent the Population of 5 cities of the province Punjab.

Cities	Population (000)
Lahore	10,355
Rawalpindi	4,765
Faisalabad	3,675
Sargodha	1,550
Multan	3,100



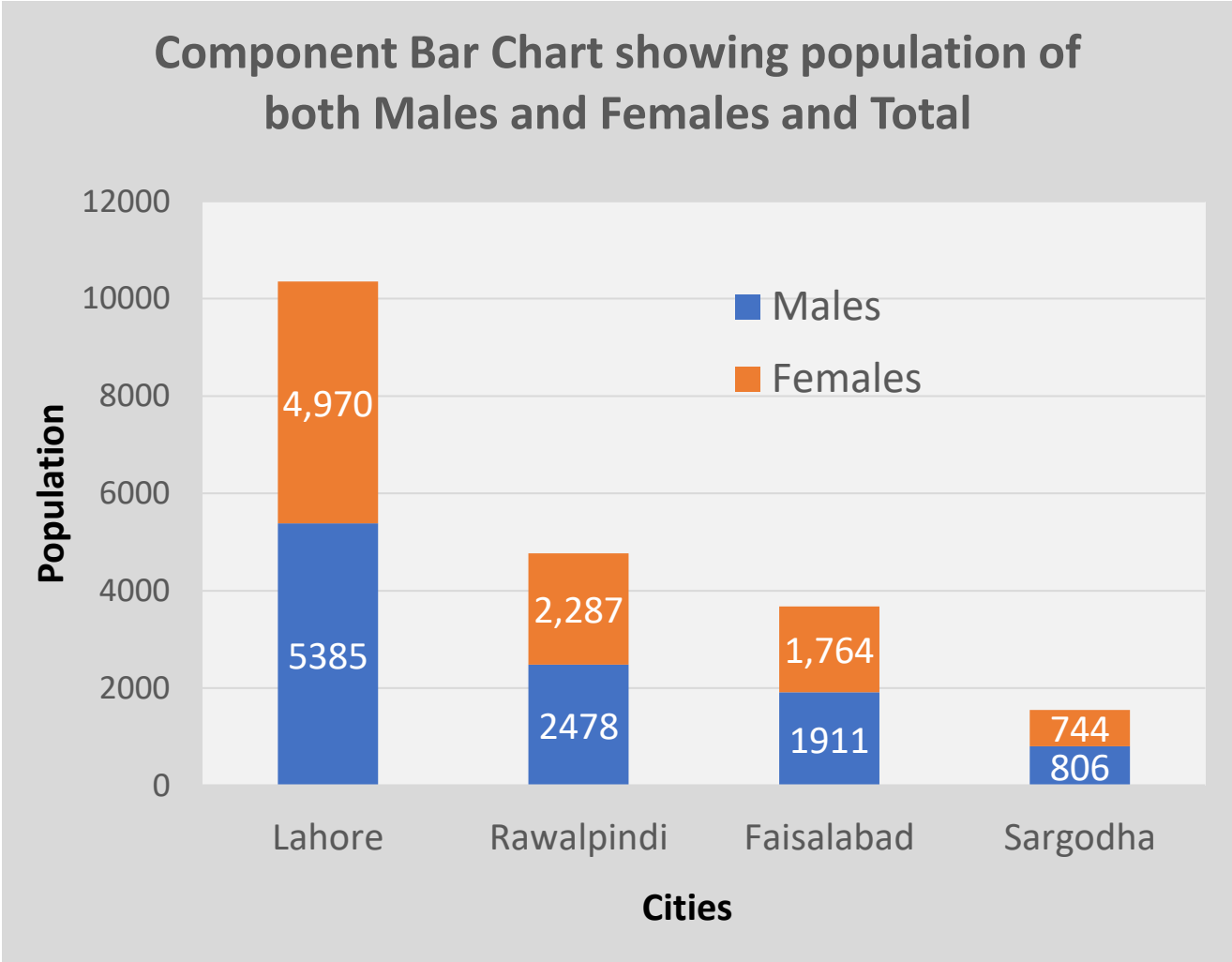
Multiple Bar Chart

Cities	Population (000)	Male	Female
Lahore	10,355	5385	4,970
Rawalpindi	4,765	2478	2,287
Faisalabad	3,675	1911	1,764
Sargodha	1,550	806	744



Component Bar Chart

Cities	Pop (000)	Male	Female
Lahore	10,355	5385	4,970
Rawalpindi	4,765	2478	2,287
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Discrete data – Frequency Distribution

Example:

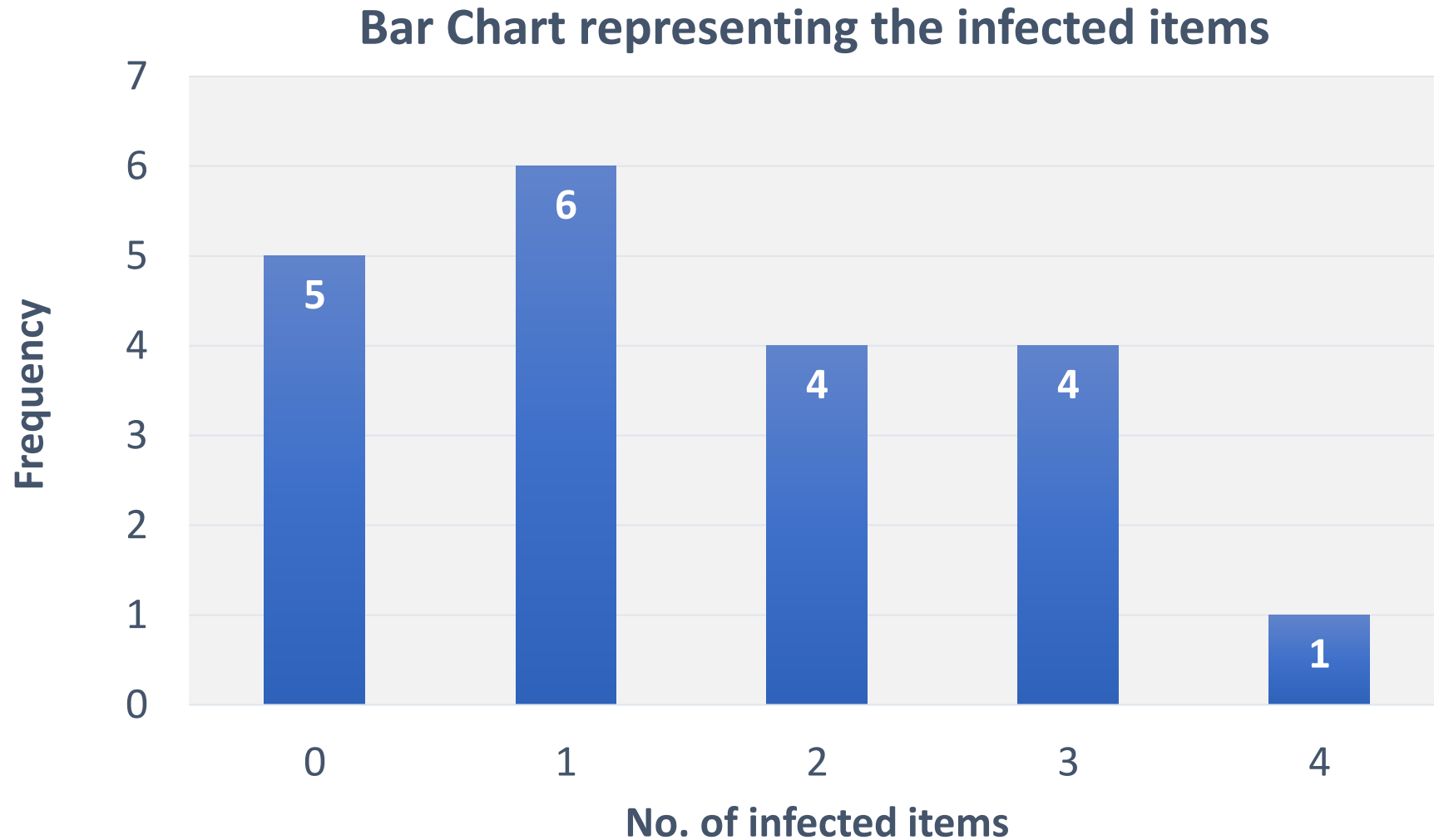
- Following data represents the number of infected plants from a sample of twenty experimental plots. Your task is to present it in tabular form.

1	2	4	3	0	1	2	3	1	1	0
2	1	0	2	3	0	0	1	3		

Discrete Frequency Distribution

No. of infected items x	Tally	Frequency f	Relative frequency
0	 	5	$5/20 = 0.25$
1	 	6	0.30
2		4	0.20
3		4	0.20
4		1	0.05
Total		20	1.00

Graphical Representation of Discrete Data

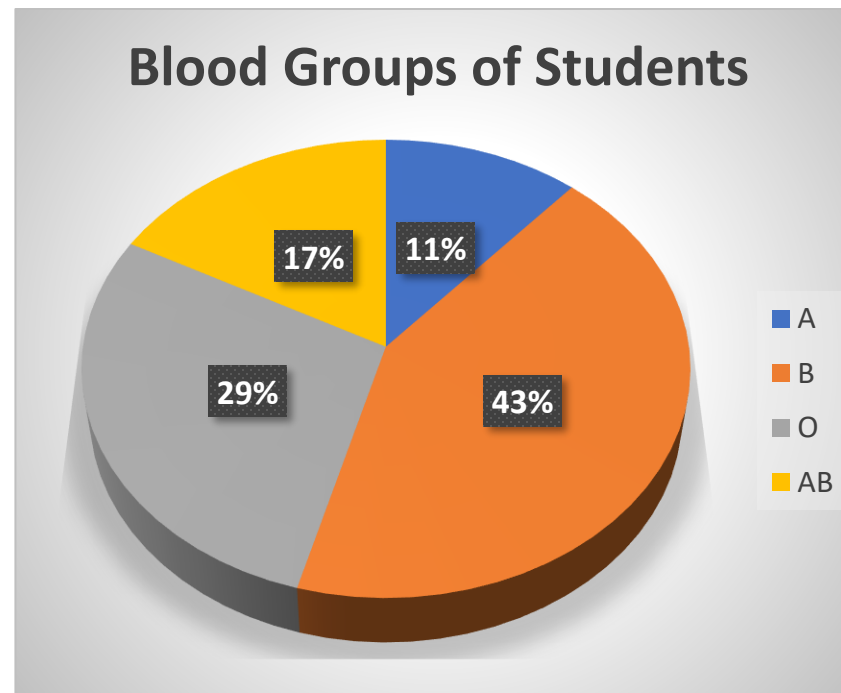


Pie Chart

- A **pie chart** is a type of graph in which a circle is divided into sectors that each represent a proportion of the whole.

Example: The blood group of 70 students were tested and the following results were obtained.

Blood Groups	No. of Students (f)
A	8
B	30
O	20
AB	12



Pie Chart

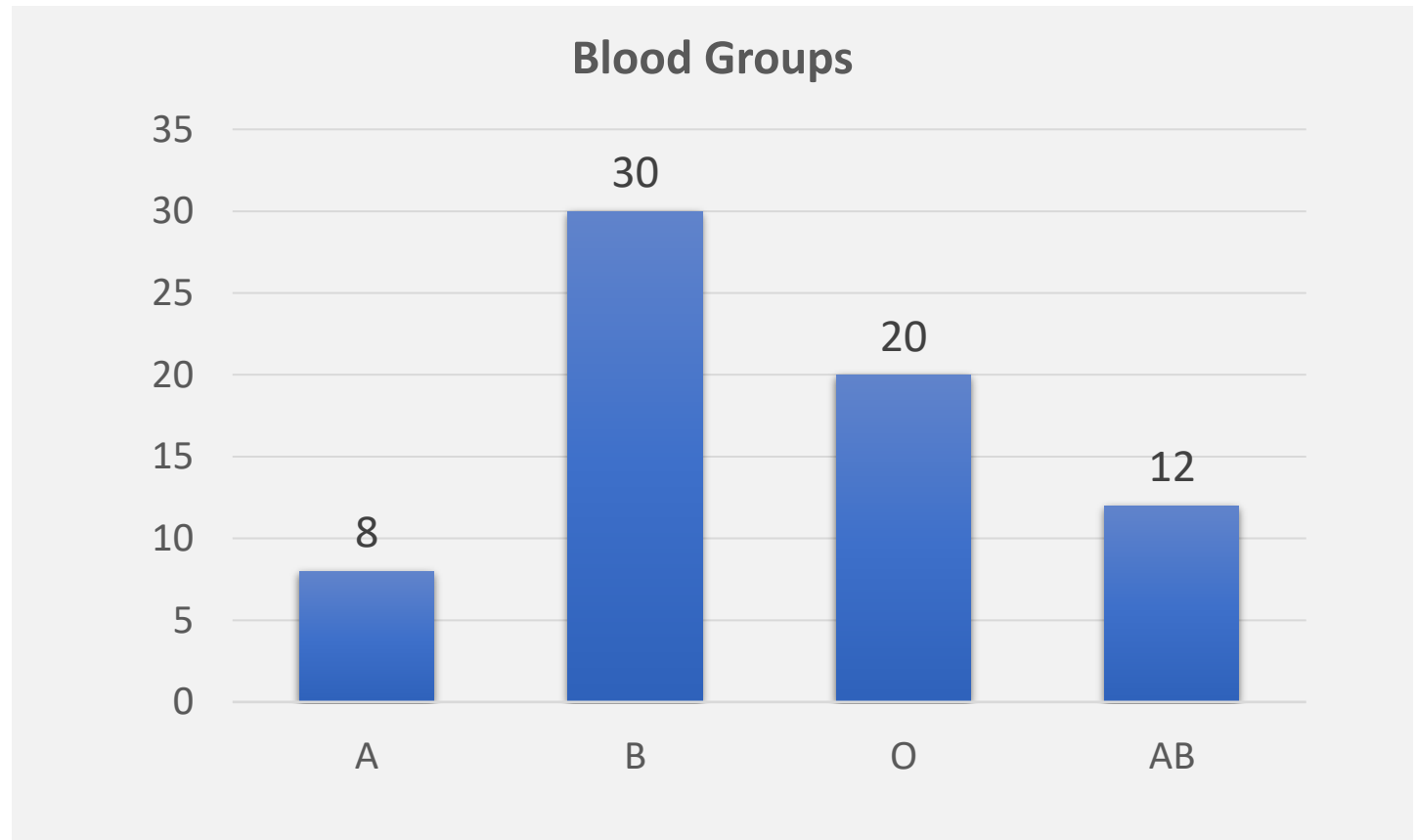
Blood Groups	No. of Students (f)	Relative frequency	Percent frequency	Angle rf x 360
A	8	$8/70 = 0.11$	$0.11 * 100 = 11$	39.6
B	30	0.43	43	154.8
O	20	0.29	29	104.4
AB	12	0.17	17	61.2
Total	70	1.00	100	360

Divide the total angle of the Circle 360 into four segments as calculated

Simple Bar Chart

- Consider the Same example of the blood group of 70 students

Blood Groups	No. of Students (f)
A	8
B	30
O	20
AB	12

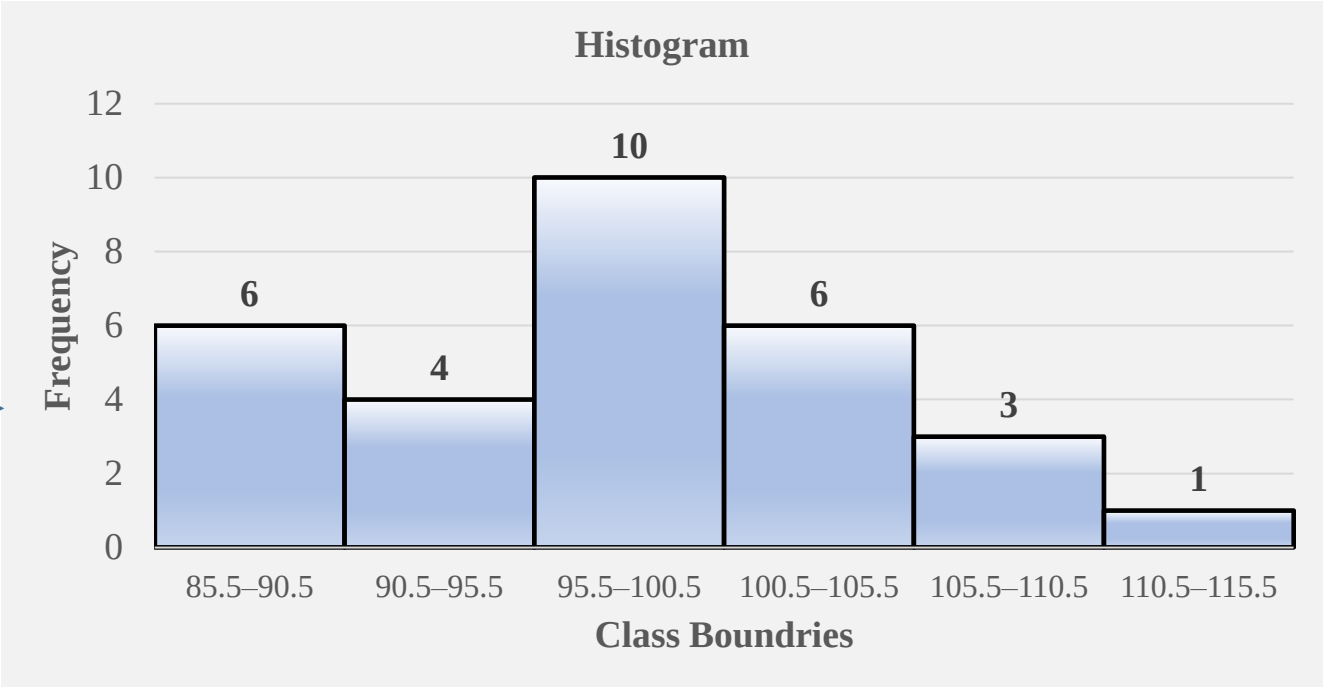


Following data represents the plant height (cm) of a sample of 30 plants.

87	91	89
88	89	91
87	92	90
98	95	97
96	100	101
96	98	99
98	100	102
99	101	105
103	107	105
106	107	112

Classes	Frequency (f)	c.f.	r.f.	% freq
86–90	6	6	0.200	20.0
91–95	4	10	0.133	13.3
96–100	10	20	0.333	33.3
101–105	6	26	0.200	20.0
106–110	3	29	0.100	10.0
111–115	1	30	0.033	3.3
Total	30		1.000	100.0

Frequency distribution & Histogram



Frequency Distribution

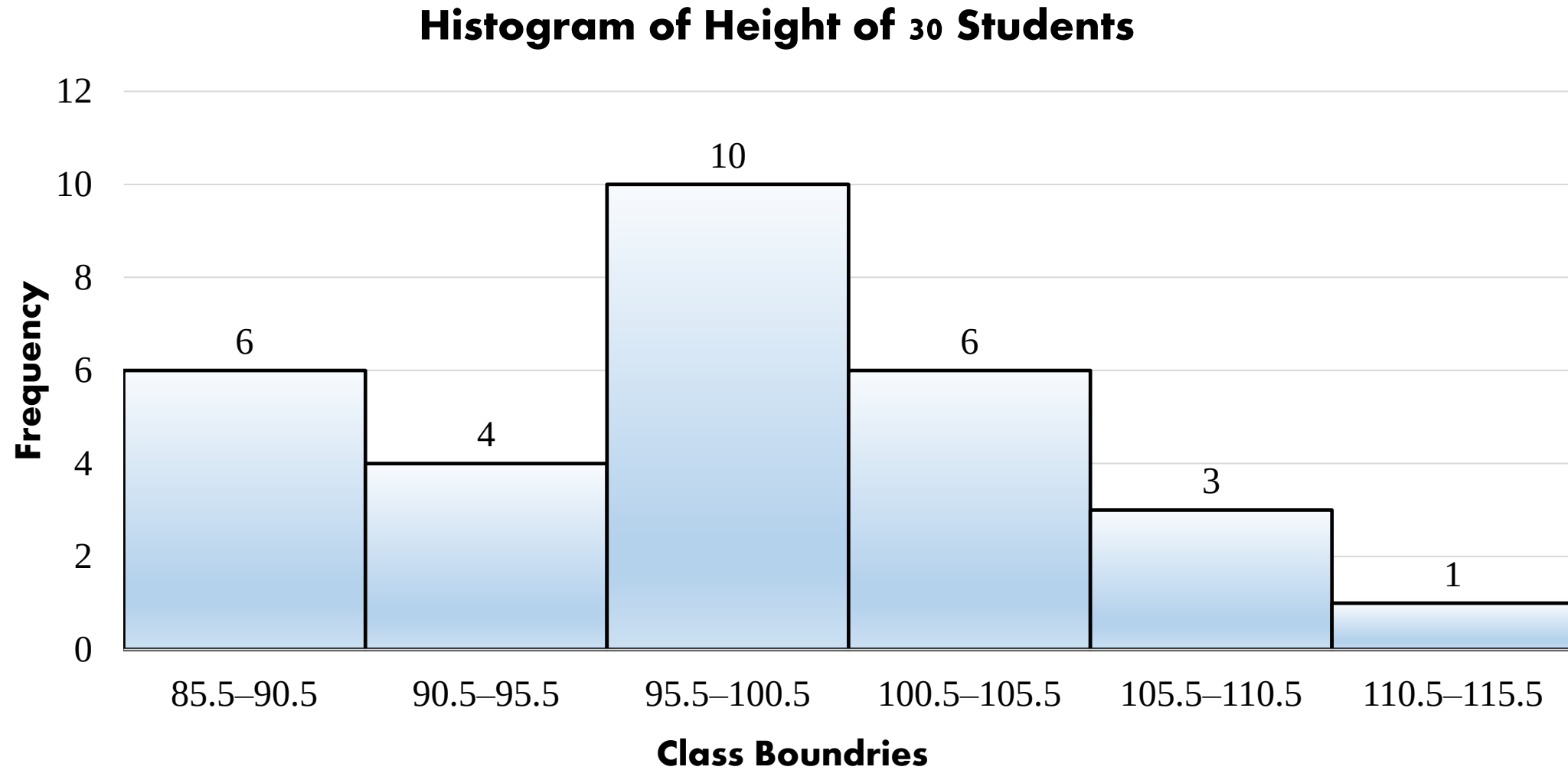
Classes	Class Boundaries	Tally	Freq (f)	c.f.	r.f.	% freq	Cumulative % freq
86–90	85.5–90.5		6	6	0.200	20.0	20.00
91–95	90.5–95.5		4	10	0.133	13.3	33.3
96–100	95.5–100.5		10	20	0.333	33.3	66.6
101–105	100.5–105.5		6	26	0.200	20.0	86.6
106–110	105.5–110.5		3	29	0.100	10.0	96.6
111–115	110.5–115.5		1	30	0.033	3.3	100.0
Total			30		1.000	100.0	

Class Boundaries

- Class Boundaries
- Subtract any Upper Class Limit from its Subsequent Lower Class limit and divide the difference with 2, you will get the Continuity correction factor
- Subtract this factor from all Lower Class Limits and add it to all Upper Class limits.

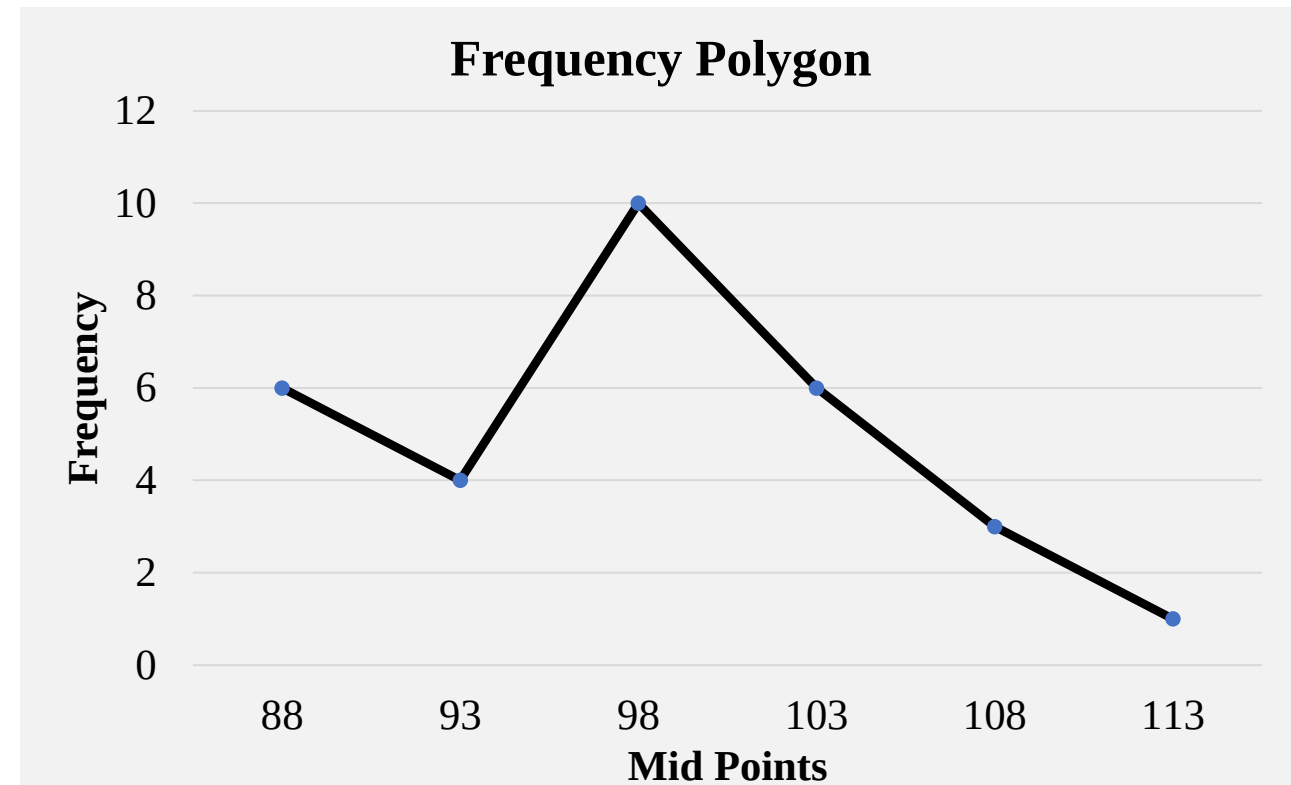
For example, $91-90 = \frac{1}{2} = 0.05$ or $96-95 = \frac{1}{2} = 0.05$

Histogram



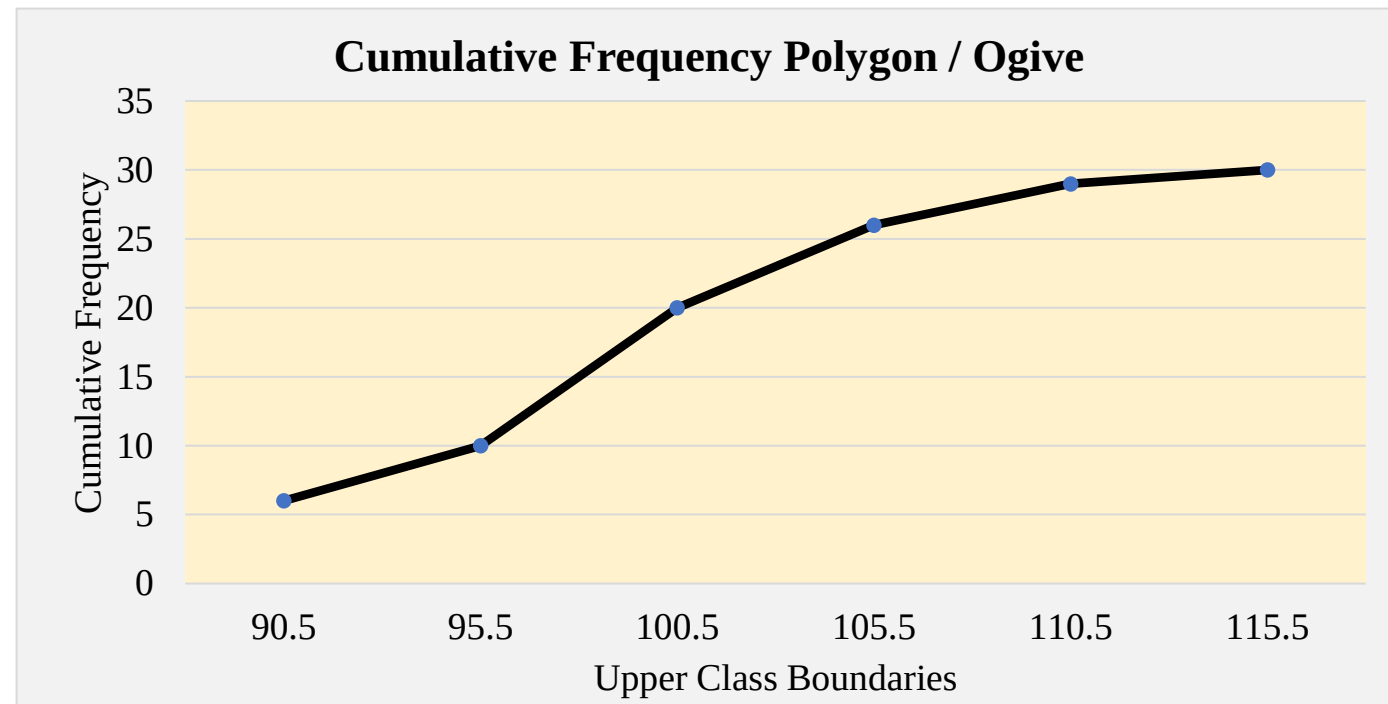
Frequency Polygon

- **Frequency** polygons are a graphical device for understanding the shapes of distributions. They serve the same purpose as histograms, but are especially helpful for comparing sets of data.
- Mid Points vs Frequency



Cumulative Frequency Polygon / Ogive

- A **cumulative frequency polygon** is a plot of the **cumulative frequency** against the upper class boundary with the points joined by a line segment.
- Upper Class Boundaries vs Cumulative Frequency



Stem & Leaf Display

- A relatively small data set can be represented by stem and leaf display.
- In addition to information on the number of observations falling in the various classes, it displays details of what those observations actually are.
- Each number in the data set is divided into two parts, a Stem and a Leaf. A stem is the leading digit(s) of each number and is used in sorting, while a leaf is the rest of the number or the trailing digit(s) and shown in display.

Example

Represent the following data by Stem and Leaf display by

(i) taking 10 unit as the width of the class

(ii) taking 5 unit as the width of the class

32 45 38 41 49 36 52 56 51 62
63 59 68

<u>Steam</u>	<u>Leaf</u>
3	2 8 6
4	5 1 9
5	2 6 1 9
6	2 3 8

<u>Steam</u>	<u>Leaf</u>	*indicate 0—4 .indicate 5—9
3*	2	
3.	8 6	
4*	1	
4.	5 9	* and . are called placeholder
5*	2 1	
5.	6 9	
6*	2 3	
6.	8	

Example

Use the data below to make a stem-and-leaf plot by taking 10 as a unit.

85	115	126	92	104
85	116	100	121	123
79	90	110	129	108
107	78	131	114	92
131	88	97	99	116
93	84	75	70	132

Stem	Leaf
7	0 5 8 9
8	4 5 5 8
9	0 2 2 3 7 9
10	0 4 7 8
11	0 4 5 6 6
12	1 3 6 9
13	1 1 2

7 | 0 5 8 9

These values are 70, 75, 78 and 79